RETAIL CUSTOMER PURCHASE PREDICTION AND PRODUCT RECOMMENDATION SYSTEM

MPOnline Internship & Certification Programs
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Internship Project Report

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ABSTRACT

The retail industry generates a large amount of customer transaction data every day. Analyzing this data helps businesses understand customer purchasing behavior and make informed marketing decisions. Predicting whether a customer is likely to make another purchase enables retailers to retain valuable customers through targeted promotions and personalized recommendations.

This project presents an intelligent Retail Customer Purchase Prediction and Product Recommendation System developed using Python, Machine Learning, and Streamlit. The system processes historical transaction records, performs data cleaning and feature engineering, and trains a Logistic Regression model to estimate the probability of a customer making another purchase within a selected prediction horizon. In addition to prediction, the application recommends products using a hybrid recommendation approach that combines purchase history, frequently bought-together products, contextual popularity, and overall product popularity.

The system provides an interactive dashboard that displays business insights such as monthly revenue, city-wise revenue, customer statistics, and product popularity. The application is designed to be user-friendly and serves as a decision-support tool for retailers to improve customer engagement and sales performance.

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CHAPTER 1

INTRODUCTION

The rapid growth of the retail industry has resulted in enormous volumes of customer transaction data. Every purchase made by a customer provides valuable information that can be analyzed to understand buying behavior, preferences, and shopping patterns. Retail businesses use this information to improve customer satisfaction, increase sales, and build long-term relationships with customers.

Customer retention is one of the most important challenges faced by retailers. Acquiring new customers is significantly more expensive than retaining existing ones. Therefore, identifying customers who are likely to make future purchases helps businesses focus their marketing efforts more effectively.

Machine Learning provides powerful techniques for analyzing historical transaction data and predicting future customer behavior. Predictive analytics enables retailers to estimate purchase probability, identify loyal customers, and recommend products that are relevant to individual customers.

The proposed system uses a Logistic Regression model to predict whether a customer is likely to make another purchase within a predefined number of days. The application also includes a recommendation engine that suggests products based on customer history and transaction patterns. The complete system is implemented using Python and deployed through an interactive Streamlit web application.

CHAPTER 2

PROBLEM STATEMENT

Retailers find it difficult to predict customer buying behavior. Develop an AI system that analyzes transaction history and predicts future purchases for personalized recommendations.

The major challenges include:

- Difficulty in identifying customers likely to return.
- Lack of personalized product recommendations.
- Inefficient marketing campaigns.
- Poor utilization of transaction history.

- Absence of intelligent decision-support tools.

The project addresses these challenges by developing a machine learning-based application capable of predicting customer purchases and recommending suitable products using historical retail transaction data.

CHAPTER 3

OBJECTIVES

The primary objectives of the project are:

- To analyze historical retail transaction data.
- To clean and preprocess customer transaction records.
- To engineer meaningful customer behavior features.
- To build a machine learning model for purchase prediction.
- To recommend products based on customer purchase history.
- To visualize retail business insights through an interactive dashboard.
- To develop an easy-to-use web application for retail decision support.

CHAPTER 4

LITERATURE SURVEY

Customer purchase prediction has become an important application of machine learning in retail analytics. Many researchers have developed predictive models that estimate customer buying behavior using transaction history, customer demographics, and purchasing patterns.

Logistic Regression is widely used for binary classification problems because of its simplicity, efficiency, and interpretability. Recommendation systems have evolved from popularity-based recommendations to hybrid recommendation techniques that combine multiple signals such as purchase history, customer similarity, and product association.

Modern retail applications integrate machine learning models with interactive dashboards, enabling businesses to visualize customer trends and make informed decisions in real time.

CHAPTER 5

SYSTEM ANALYSIS

Existing System

Traditional retail management systems primarily generate sales reports and inventory information. These systems have several limitations:

- No prediction of future customer purchases.
- No personalized recommendations.
- Limited customer behavior analysis.
- Manual decision-making.
- Static reporting system.

Proposed System

The proposed Retail Customer Purchase Prediction and Product Recommendation System overcomes these limitations by integrating machine learning with retail analytics.

The system performs:

- Data validation
- Data preprocessing
- Feature engineering
- Machine learning prediction
- Product recommendation
- Interactive visualization

The application allows users to train the prediction model, analyze customer behavior, and generate personalized product recommendations using historical transaction data.

CHAPTER 6

SYSTEM DESIGN

Software Requirements

- Python 3.x
- Streamlit
- Pandas
- Scikit-learn
- Plotly
- Joblib
- Visual Studio Code

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: 2 GB Free Space
- Operating System: Windows 10/11

Folder Structure

```
Major_Project/
        ├── .venv/
        ├── __pycache__/
        ├── models/
        │   └── purchase_predictor.joblib
        │
        ├── src/
        │   ├── __init__.py
        |   ├──  __pycache__
        │   └── retail_predictor.py
        │
        ├── app.py
        ├── train_model.py
        ├── Retail_Transactions_Dataset.csv
        ├── requirements.txt
        ├── README.md
        ├── PROJECT_REPORT.md
        ├── run_project.bat
```

CHAPTER 7

METHODOLOGY

The project follows the following workflow:

Step 1: Data Collection

The application loads the retail transaction dataset stored in CSV format.

Step 2: Data Cleaning

The dataset is validated by:

- Removing invalid records.
- Handling missing values.
- Parsing transaction dates.
- Converting numerical values.
- Processing product information.

Step 3: Feature Engineering

The application creates customer behavior features such as:

- Number of previous purchases.
- Average spending.
- Average number of purchased items.
- Discount usage rate.
- Days since previous purchase.
- Number of unique products purchased.

These features help improve prediction accuracy.

Step 4: Model Training

The project uses **Logistic Regression** for binary classification.

The target variable is:

Will Purchase Within Prediction Horizon

Possible values:

- 1 → Customer will purchase again.
- 0 → Customer will not purchase again.

Step 5: Prediction

The trained model estimates the probability that a customer will make another purchase within the selected prediction horizon.

Step 6: Product Recommendation

The recommendation engine combines multiple recommendation strategies:

- Customer purchase history
- Frequently bought together products
- Contextual popularity
- Overall product popularity

This hybrid approach provides personalized recommendations.

CHAPTER 8

IMPLEMENTATION

The project consists of three major modules.

User Interface Module

Developed using Streamlit.

Functions:

- Upload dataset
- Train model
- Customer selection
- Purchase prediction
- Product recommendation
- Dashboard visualization

Machine Learning Module

Responsible for:

- Feature engineering
- Model training
- Model evaluation
- Prediction

The trained model is stored as a Joblib file for future use.

Recommendation Module

Generates personalized product recommendations based on customer history and transaction patterns.

CHAPTER 9

RESULTS

The developed application successfully predicts customer purchase probability using historical transaction data.

The dashboard displays:

- Number of valid transactions
- Total customers
- Total revenue
- Average order value
- Monthly revenue trend
- Revenue by city
- Frequently purchased products

The customer prediction module provides:

- Purchase probability
- Risk level
- Number of purchases
- Total customer spending

The recommendation module displays personalized product recommendations for the selected customer.

System Dashboard

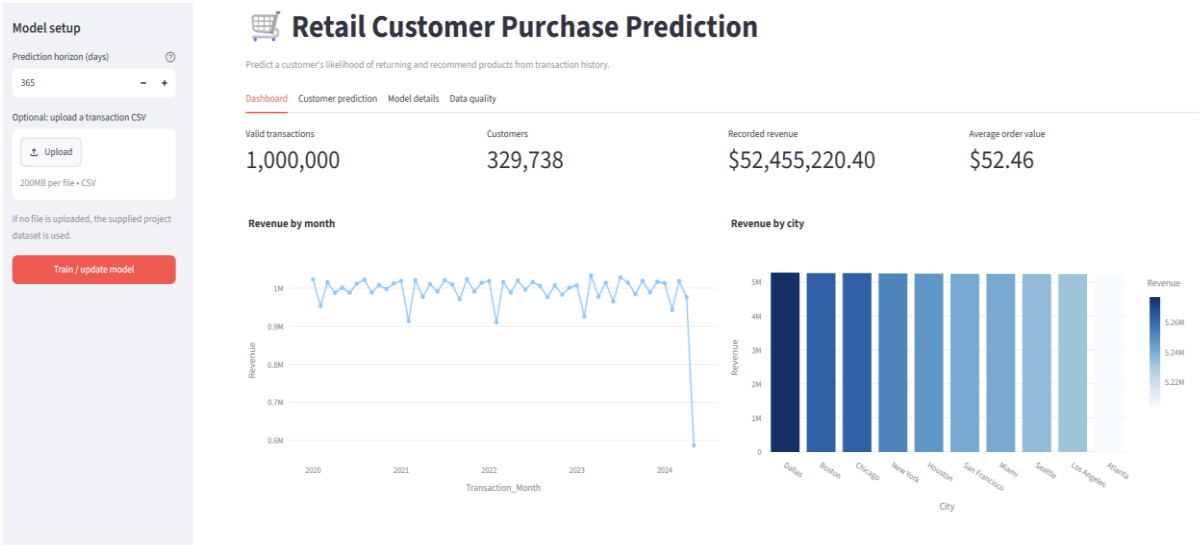


Figure: Dashboard displaying key business metrics, monthly revenue trend, revenue by city, and model setup options. The dashboard provides an overview of retail transaction data and enables users to train or update the prediction model.

Customer Purchase Prediction

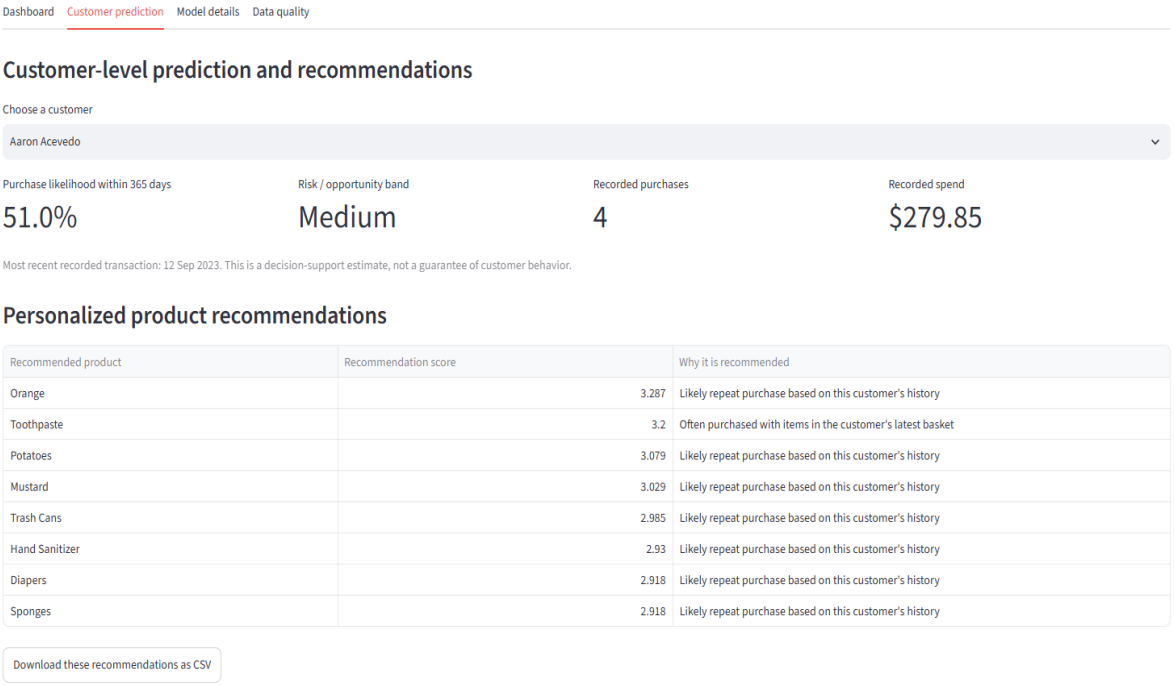
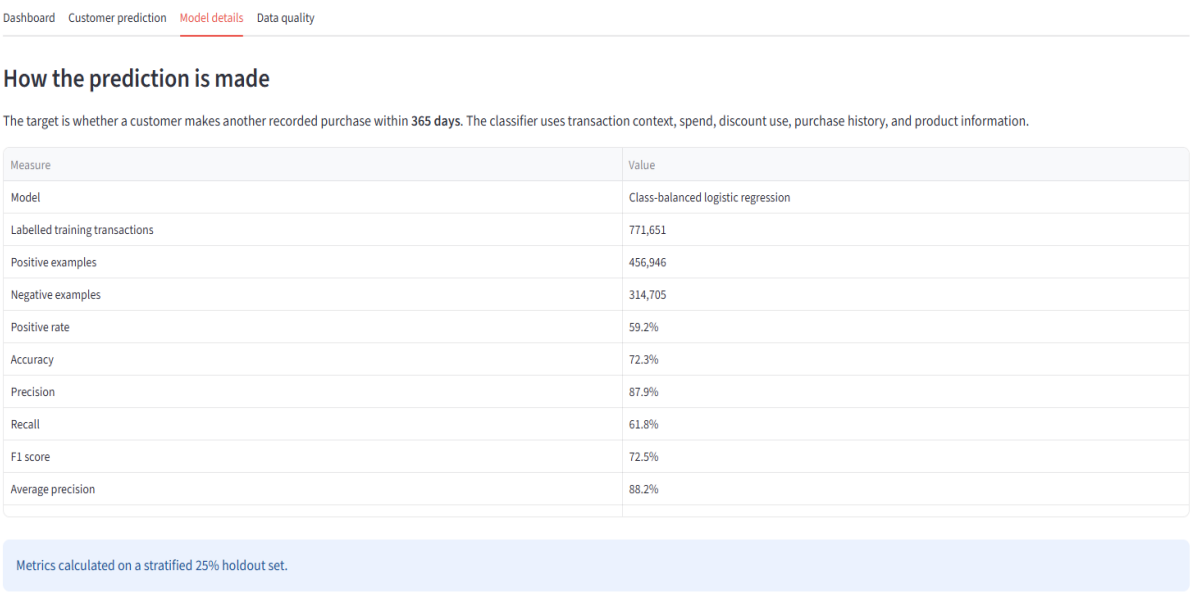


Figure: Customer prediction interface showing purchase probability, customer spending information, risk category, and personalized product recommendations generated by the hybrid recommendation engine.

Model Evaluation



Dashboard Customer prediction Model details Data quality

How the prediction is made

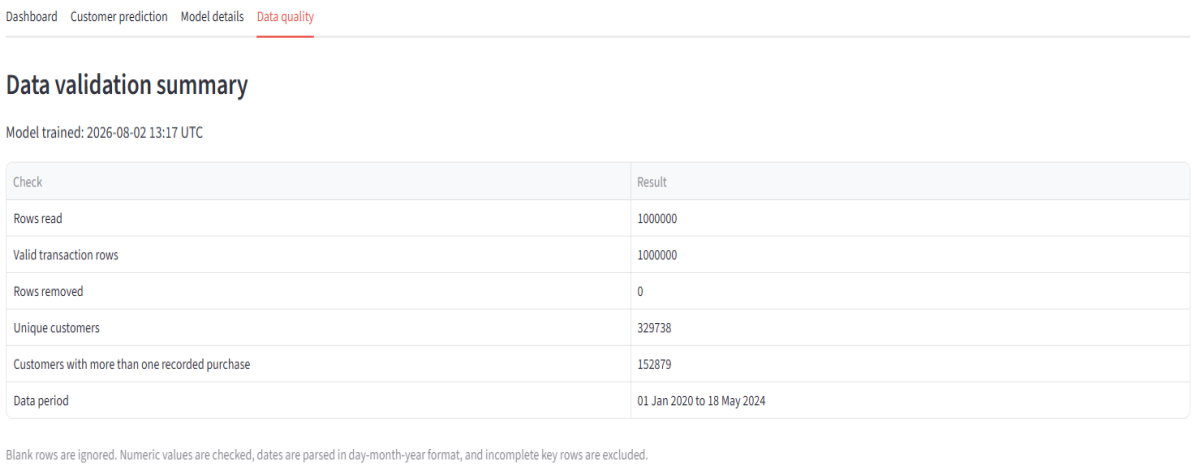
The target is whether a customer makes another recorded purchase within **365 days**. The classifier uses transaction context, spend, discount use, purchase history, and product information.

Measure	Value
Model	Class-balanced logistic regression
Labelled training transactions	771,651
Positive examples	456,946
Negative examples	314,705
Positive rate	59.2%
Accuracy	72.3%
Precision	87.9%
Recall	61.8%
F1 score	72.5%
Average precision	88.2%

Metrics calculated on a stratified 25% holdout set.

Figure: Model evaluation page displaying the machine learning algorithm used along with performance metrics including Accuracy, Precision, Recall, F1-Score, Average Precision, and training statistics.

Data Validation



Dashboard Customer prediction Model details Data quality

Data validation summary

Model trained: 2026-08-02 13:17 UTC

Check	Result
Rows read	1000000
Valid transaction rows	1000000
Rows removed	0
Unique customers	329738
Customers with more than one recorded purchase	152879
Data period	01 Jan 2020 to 18 May 2024

Blank rows are ignored. Numeric values are checked, dates are parsed in day-month-year format, and incomplete key rows are excluded.

Figure: Data validation summary showing the number of processed records, valid transactions, unique customers, repeat customers, and the transaction period used for model training.

CHAPTER 10

ADVANTAGES

- Easy-to-use web interface.
- Automatic model training.
- Customer purchase prediction.
- Hybrid recommendation system.
- Interactive business dashboard.
- Data validation and preprocessing.
- Reusable trained model.
- Personalized customer insights.

CHAPTER 11

LIMITATIONS

- Prediction quality depends on historical data.
- Limited repeat customer transactions reduce prediction confidence.
- Only one machine learning algorithm is implemented.
- The system uses offline CSV data and does not process real-time transactions.

CHAPTER 12

FUTURE SCOPE

The project can be enhanced by:

- Using larger retail datasets.
- Integrating real-time databases.
- Implementing advanced machine learning algorithms such as Random Forest, XGBoost, or Deep Learning models.
- Deploying the application on cloud platforms.
- Adding customer segmentation and sales forecasting.
- Integrating online retail platforms for real-time recommendations.

CHAPTER 13

CONCLUSION

The Retail Customer Purchase Prediction and Product Recommendation System demonstrates how machine learning can improve retail decision-making by analyzing historical transaction data. The application successfully preprocesses retail data, extracts meaningful customer features, predicts future purchasing behavior using Logistic Regression, and generates personalized product recommendations.

The interactive dashboard enables retailers to visualize important business metrics, while the recommendation engine supports customer engagement through relevant product suggestions. Although the current implementation is limited by the availability of repeat-customer data, it provides a strong foundation for future enhancements and demonstrates the practical application of predictive analytics in the retail domain.

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